

Activity 1.1.6 Maximizing Power

PLTW

MAXIMIZING POWER

- ≡ Motor Performance
- ≡ Maximizing Motor Power
- ≡ No-Load Performance
- ≡ Stall Performance
- ≡ Motor Power
- ≡ Motor Efficiency
- ≡ Conclusion

Motor Performance

GOALS	MATERIALS	RESOURCES
• Take measurements of motors. • Create and interpret characteristic curves that provide data on maximum power and efficiency.		
GOALS	MATERIALS	RESOURCES
• POE Gearbox Motor Winch from Activity 1.1.3 Mechanical System Efficiency • Timing device to measure time to 0.01-second precision • Vernier® Go Direct® Force and Acceleration Sensor		

- Vernier Graphical Analysis® application

GOALS	MATERIALS	RESOURCES
		• <u>POE Gearbox Motor Winch Construction Guide</u> • <u>Multimeter Operation</u> • <u>Vernier Graphical Analysis®</u> • <u>Vernier® Sensors and Probes</u> • <u>PLTW Engineering Formula Sheet</u> • <u>Motor Power Excel Spreadsheet</u> • <u>Motor Power Google Sheets Spreadsheet</u> • <u>Activity 1.1.6 Maximizing Power (Downloadable PDF)</u>

When designing new systems, an engineer often has access to limited resources. For example, to design a laptop or smartphone, an engineering team works to make the battery last as long as possible which means battery size is a constraint. A vibrating smartphone uses a motor with an off-center load, and it needs to be designed with a gear train that delivers the most [power](#) or highest [efficiency](#) but also fits within the existing device.



In this activity, you will learn how to take the measurements of a motor. You will also create characteristic curves that you can use to select a motor that delivers maximum power or maximum efficiency for a given task.

To develop specifications for a motor, you need four measurements: power, efficiency, current, and speed. These four measurements describe the most important properties of a particular motor at a given voltage and help an engineer develop and analyze the curves that characterize or describe the technical specifications of a motor like the one shown in Figure 1.

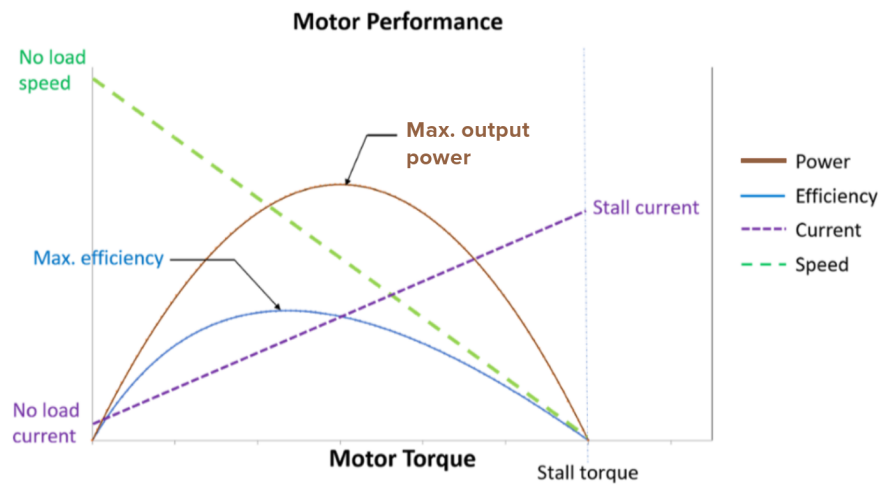


Figure 1. Motor Characteristics Curves

Maximizing Motor Power

Motors are designed to convert electrical energy into mechanical energy. You have likely noticed motor power when driving up a steep hill or when towing a heavy load. As you saw in Activity 1.1.3 Mechanical System Efficiency, whenever motors are used, the goal is to be as efficient as possible. When adding loads to a motor, torque and speed need to be examined to maximize the amount of power.

Speed Depends on Load Torque

When a motor is loaded with no torque, it spins at no-load speed. The lowercase Greek letter omega (ω) is the symbol used for angular speed.

$$\omega_0 = \text{no-load speed}$$

When a motor is loaded with the stall torque or more, the motor will stop. The lower-case Greek letter tau (τ) is the symbol used for torque.

$$\tau_{\text{stall}} = \text{stall torque}$$

Speed vs. Torque Is Linear

Speed has a linear dependence on torque as shown in Figure 2.

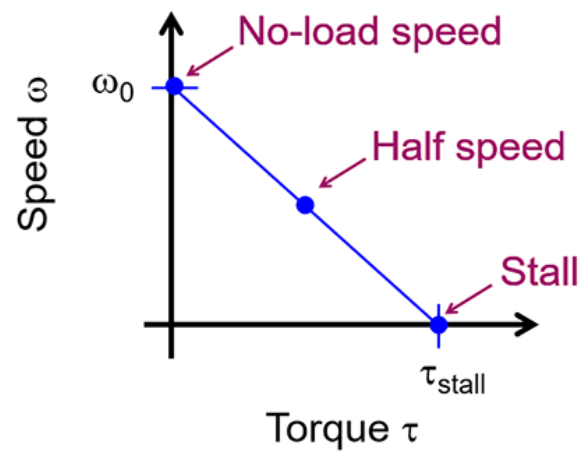


Figure 2. Speed vs. Torque graph

Algebra can help you create a formula to show this relationship.

- Formula for a line: $y = mx + b$
- y = speed (ω)
- m = slope = rise/run

$$\frac{\text{-no-load speed}}{\text{stall torque}} = \frac{-\omega_0}{\tau_{\text{stall}}}$$

- x = torque (τ)
- b = no-load speed

Combining all of these variables, we can create the formula for speed.

$$\omega = \frac{-\omega_0}{\tau_{\text{stall}}} \cdot \tau + \omega_0$$

Power

Power is the rate at which work (W) is performed in a given time (t).

The following relationship is used when relating power to torque and angular speed.

$$P = \tau \cdot \omega$$

Knowing this relationship gives us the graph shown in Figure 3.

Power = Torque x Speed

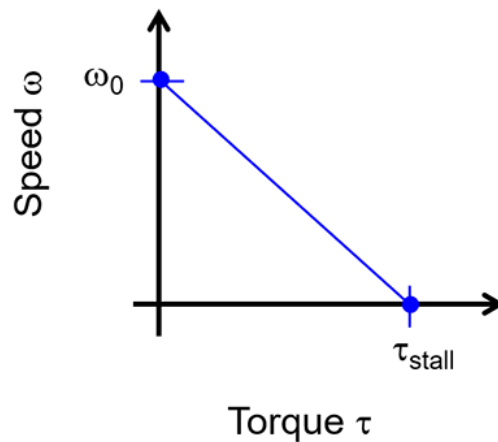


Figure 3. The relationship between torque and speed

You may notice that there is no power when torque is 0. You may also notice that when speed is 0, there is also no power. This is because power

= torque x speed, and no value for either torque or speed will give us 0 power.

Maximum Power

Similar to Activity 1.1.3 Mechanical System Efficiency, we want to get the most amount of power in order to efficiently use a motor. So that naturally begs the question, “How do I know when I am using maximum power?”.

Maximum Power at Half Stall Torque

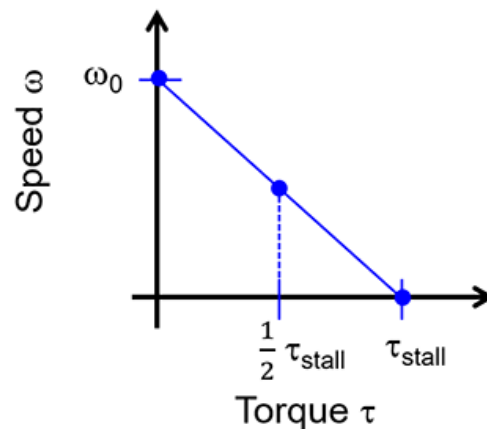


Figure 4. Maximum Power

To solve for maximum power start with the relationship between speed and torque:

$$\omega = \frac{-\omega_0}{\tau_{\text{stall}}} \cdot \tau + \omega_0$$

Then, multiplying both sides of the equation by torque:

$$\tau \cdot \omega = \left[\frac{-\omega_0}{\tau_{\text{stall}}} \cdot \tau + \omega_0 \right] \cdot \tau$$

And simplifying the equation:

$$P = \frac{-\omega_0}{\tau_{\text{stall}}} \cdot \tau^2 + \omega_0 \tau$$

Examine Figure 5 to see the relationship between power and torque.

Maximum Power at Half Stall Torque

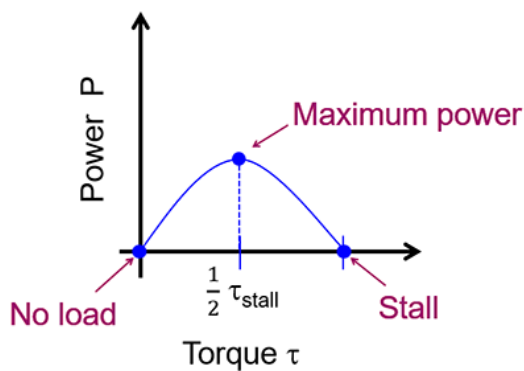


Figure 5. Power vs. Torque

Review Example Problem: Part 1 and Part 2.

EXAMPLE PROBLEM: PART 1

EXAMPLE PROBLEM: PART 2

The VEX motor in the figure is stalled. Find the stall torque and maximum power that this motor can provide.

To begin, solve for the torque when the motor is stalled:

$$\tau_{\text{stall}} = d \cdot F_{\perp}$$

$$\tau_{\text{stall}} = (3 \text{ in})(1.4 \text{ lb})$$

$$\tau_{\text{stall}} = 4.2 \text{ lb} \cdot \text{in}$$

Now, you can find the point at which the motor has maximum power:

$$\tau_{\text{maxPower}} = \frac{1}{2} \tau_{\text{stall}}$$

$$\tau_{\text{maxPower}} = \frac{1}{2} (4.2 \text{ lb} \cdot \text{in}) = 2.1 \text{ lb} \cdot \text{in}$$



EXAMPLE PROBLEM: PART 1

EXAMPLE PROBLEM: PART 2

Now, examine the same motor as in Part 1 (Figure 6). It is being used to power a winch with a drum with a radius of 2 inches and lifting .2 pound. What torque is the motor applying?

$$\tau_{\text{out}} = d \cdot F_{\perp}$$

$$\tau_{\text{out}} = (2 \text{ in})(0.2 \text{ lb})$$

$$\tau_{\text{out}} = 0.4 \text{ lb} \cdot \text{in}$$

To improve the efficiency of this motor, a gear could be added. What gear ratio would make the motor deliver maximum power?

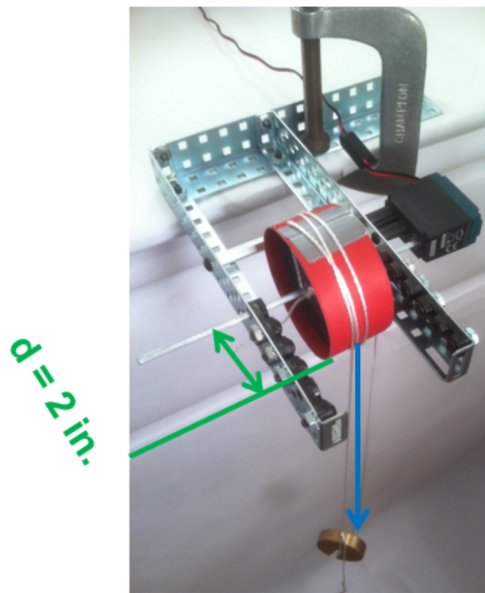
$$\text{Gear Ratio} = \frac{\tau_{\text{out}}}{\tau_{\text{in}}}$$

$$\text{Gear Ratio} = \frac{0.4 \text{ in} \cdot \text{lb}}{2.1 \text{ in} \cdot \text{lb}}$$

$$\text{Gear Ratio} = 0.19$$

$$\text{Gear Ratio} = \frac{12 \text{ teeth}}{60 \text{ teeth}}$$

Figure 7 shows the winch geared with a 60-tooth gear driving a 12-tooth gear. The motor is delivering half its stall torque, for maximum power.



$F = 0.2 \text{ lbs}$

Figure 6. Assembly Before Design Optimization

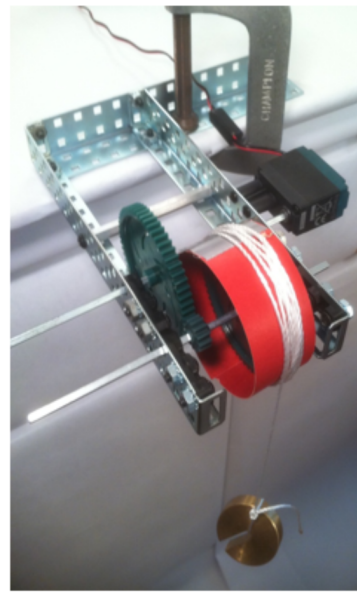


Figure 7. Assembly After Design Optimization

No-Load Performance

1

Add a Vernier® Go Direct® Force and Acceleration Sensor to the POE Gearbox Motor Winch from Activity 1.1.3 Mechanical System Efficiency as shown in Figure 8. If you need to rebuild the Gearbox Motor Winch, review the [**POE Gearbox Motor Winch Construction Guide**](#).

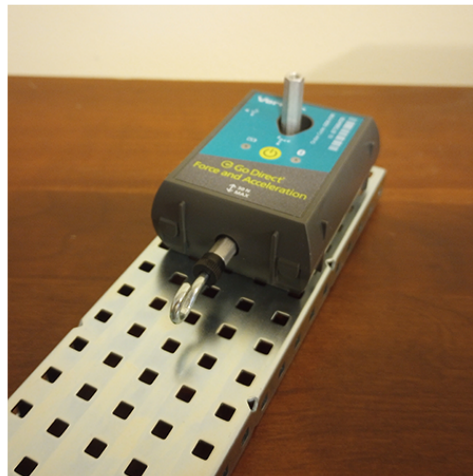


Figure 8. Force and Acceleration Sensor

2

Prepare the winch cable to make measurements:

- Unspool the winch cable.

- Pass the loop at the end of the cable through the spokes of the 4-inch wheel as shown in Figure 9.
- Loop the cable back through the end that is attached to the winch drum as shown in Figure 10. The cable loop must be in the center of the wheel circumference.
- As shown in Figure 11, secure the end of the cable onto the wheel to avoid interference with wheel movement:
 - Hold light tension on the end of the cable as you turn the top of the wheel toward the force sensor.
 - Adjust the cable alignment so that the cable is wound around the center of the wheel circumference.
 - Tape the end of the cable onto the face of the wheel.

NOTE: Figure 11 shows blue tape to improve the visibility of the image. You can use any tape to secure the end of the cable.

- Add a doubled-over piece of tape to the edge of the wheel as shown in Figure 12. This tape will strike your finger as the wheel turns to help you count the revolutions.

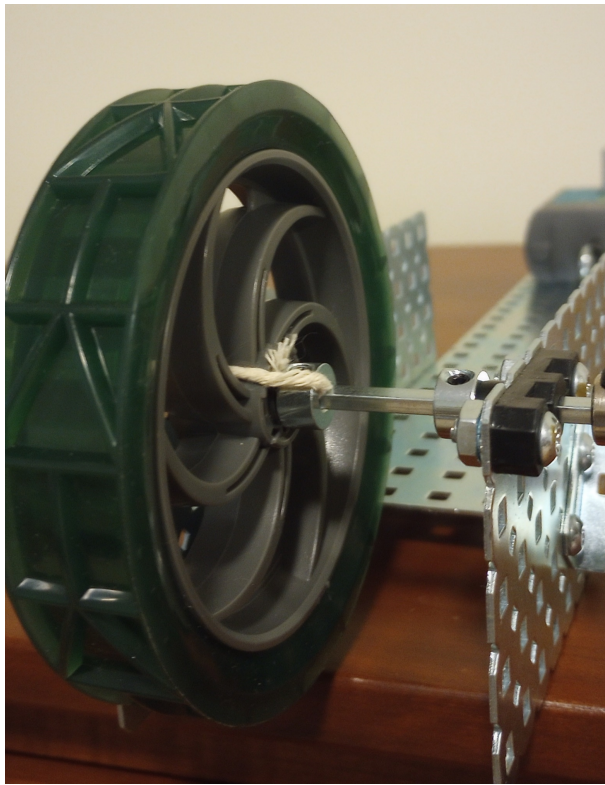


Figure 9. Cable Passed Through Spokes

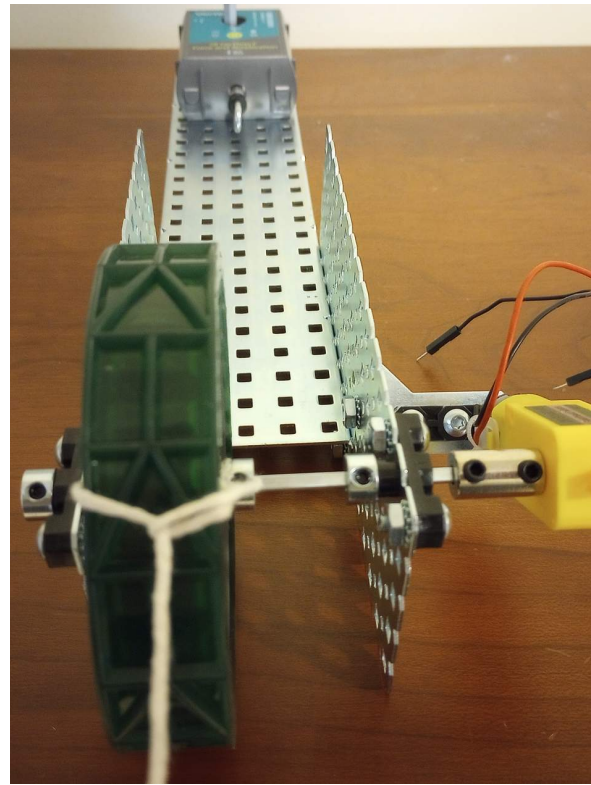
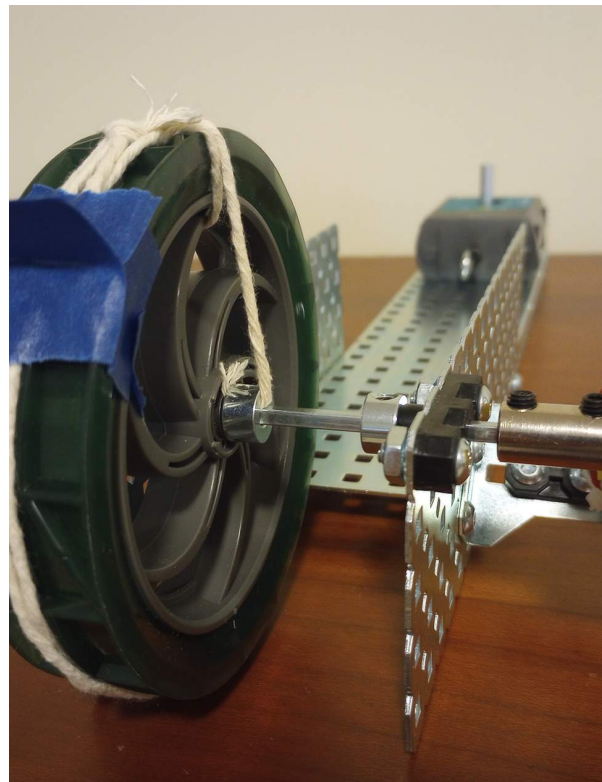
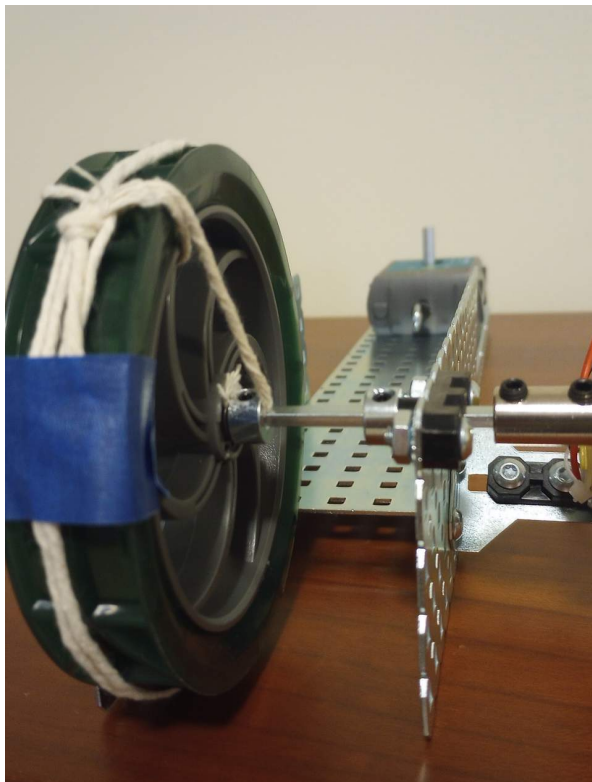


Figure 10. Cable Looped on Wheel Edge



3

Measure the no-load speed of the motor:

- Turn on the power supply.
- Adjust the voltage knob to 6 V.
- Position your finger so that the doubled-over tape strikes your finger each time the wheel turns so you can count the revolutions.
- Touch and hold the red lead from the power supply to the red multimeter test probe.
- Touch and hold the black multimeter test probe to the red lead from the motor to start the motor turning, as shown in Figure 13. For help using a multimeter, review [Multimeter Operation](#).
- Start the stopwatch when you see the tape reach and strike your finger.
- Record the voltage and the current displayed on the multimeter.
- Stop the stopwatch after the winch turns 20 times. Record the elapsed time in your PLTW Engineering Notebook.
- Repeat the series of instructions above two more times.
- Turn off the power supply.

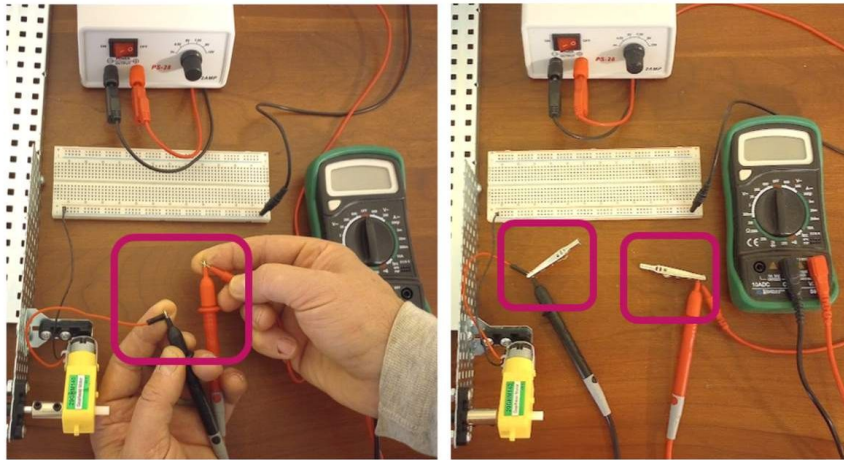


Figure 13. Electrical Connections

NOTE: A recommended practice is to measure multiple trials, and then calculate the average for each measurement. If any of your time measurements seem drastically different from the others, consider it an outlier and do not include it in the mean.

- 4
- Recreate the table below in your PLTW Engineering Notebook to record data.

Measurement	Trial 1	Trial 2	Trial 3	Average
n (0.0) Number of rotations				

t (0.00 s) Time for n rotations				
$V_{\text{no-load}}$ (0.0 V)				
$I_{\text{no-load}}$ (0.00 A)				

5

Calculate the no-load speed, which is the fastest a motor will spin at a voltage. Use the formula in the table below.

Formulas	Substitute/Solve	Final Answer
No-Load Speed (0.00 rotations / s) $\omega_{\text{no-load}} = \frac{\text{number of rotations}}{\text{time}}$		
Convert to no-load speed in radians per second where 2π radians = 1 rotation		

Stall Performance

6

Remove the tape that secured the cable to the winch wheel.

7

Hold light tension on the end of the cable as you hook the loop onto the force sensor hook as shown in Figure 14.

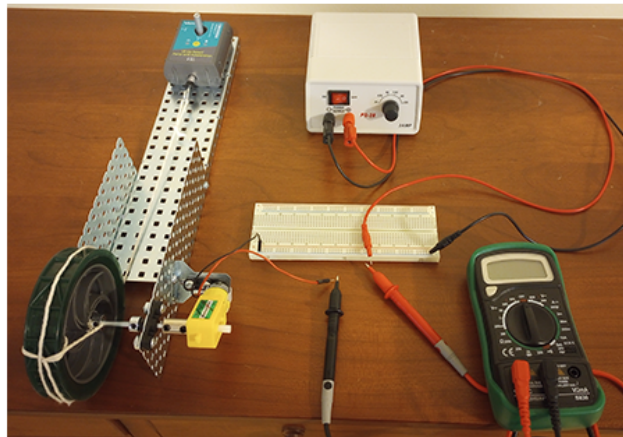


Figure 14. Stall Measurement Configuration.

This setup in Figure 14 shows a Dual-Range Force Sensor. The setup is the same for the Vernier® Go Direct® Force and Acceleration Sensor.

NOTE: The cable should contact the wheel at the bottom of the circumference, not the top.

8

To set up the force sensor, use the following steps:

Vernier Sensors and Graphical Analysis: For more details and videos about connecting sensors and collecting, saving, and printing data, refer to the [Vernier Graphical Analysis®](#) and [Vernier® Sensors and Probes](#) resources.

- Using the USB cable, plug the micro-USB connector into the Go Direct Force and Acceleration sensor and the USB connector into your computer.
- Launch Graphical Analysis on your computer and select **Sensor Data Collection**.
- On the Sensors box, select **Connect** and **DONE**.



9

Measure the stall performance of the motor:

- Turn on the power supply.
- Adjust the Voltage knob to 6 V.



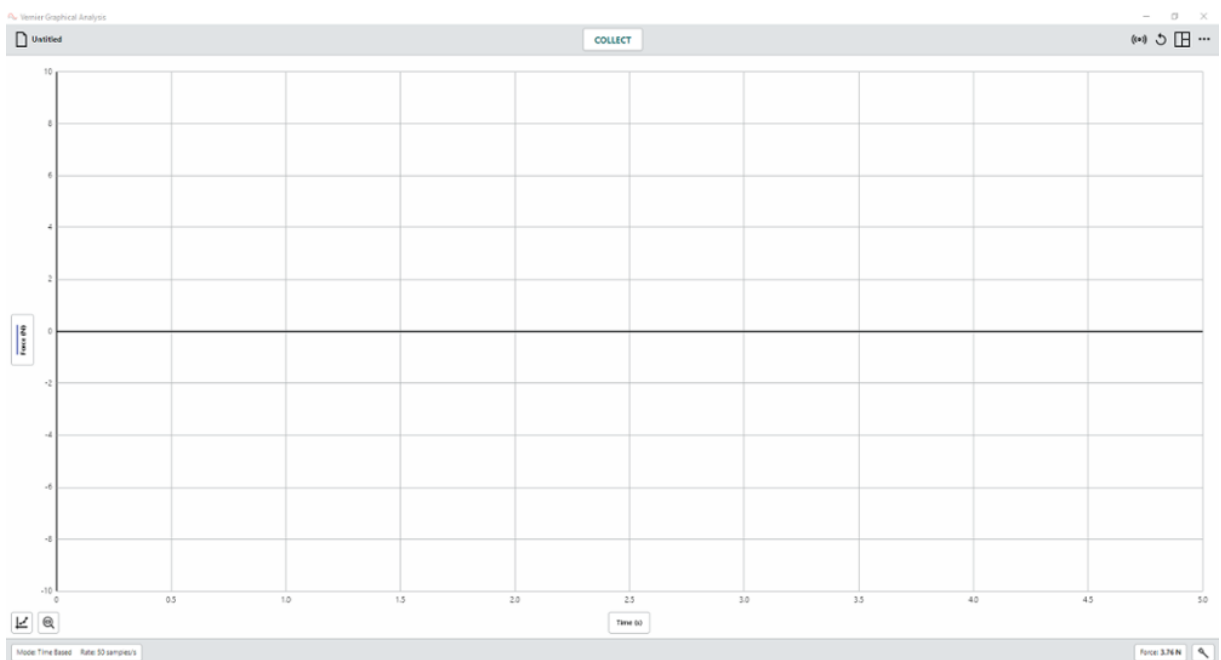
Safety Reminder: Before you proceed, read this warning.

The Gearbox Motor has no overcurrent protection, so it can be damaged by too much current. The current heats up the motor coils and melts the insulation, short-circuiting the coils to each other. Good practice is to take this measurement quickly and then turn off the power to the motor. Let the Gearbox Motor cool for about 10 seconds before you measure again.

Prepare Graphical Analysis to record force data in the following steps.

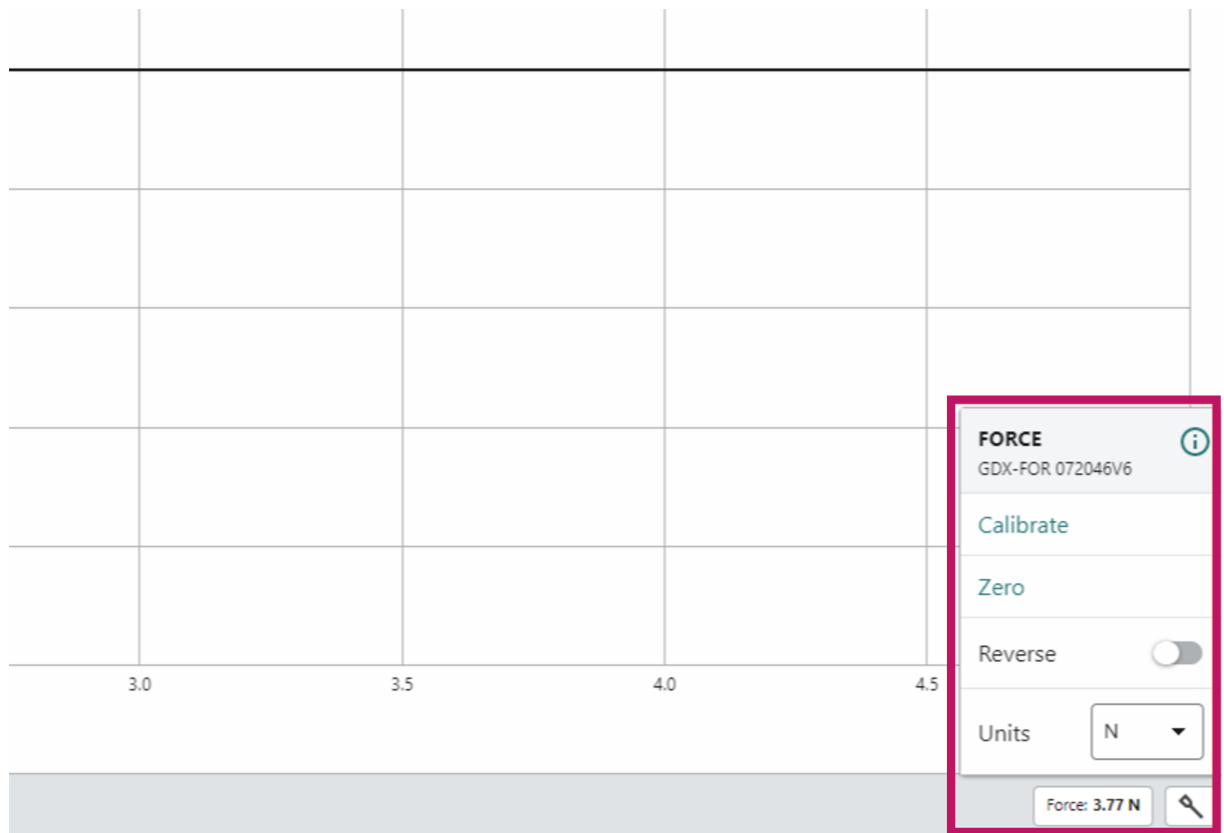
Step 1

With the Go Direct Force and Acceleration Sensor connected to Graphical Analysis, you should see a graph ready to collect data.



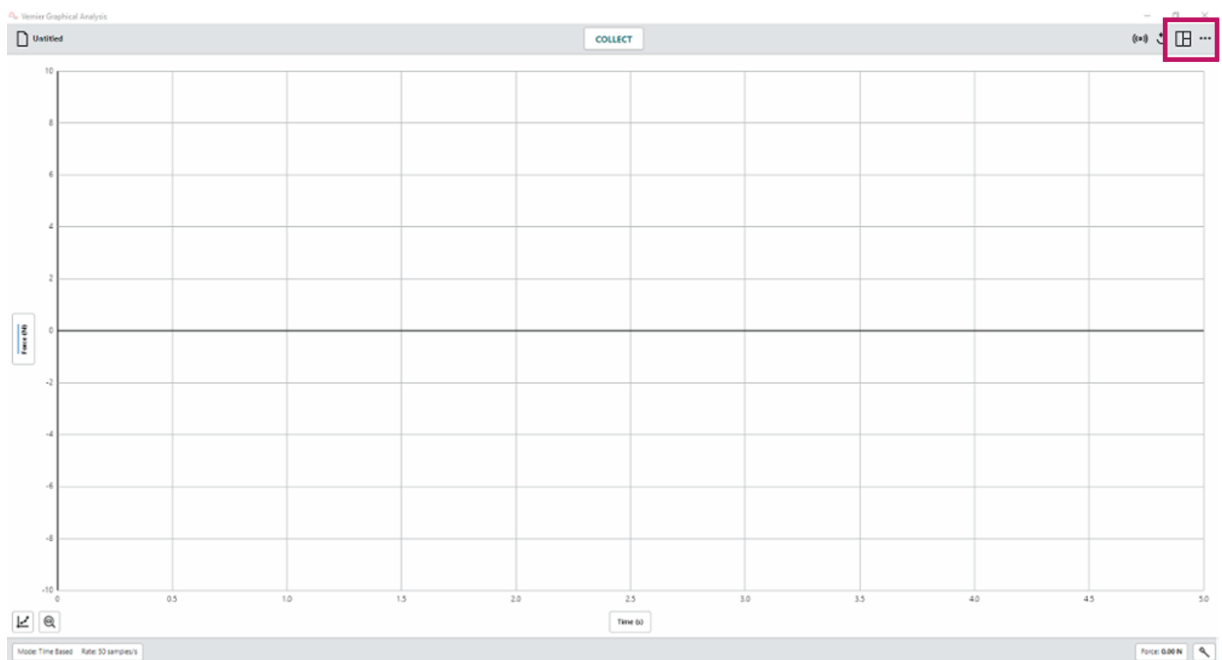
Step 2

If your sensor value is not 0.0 N (0 Newtons), you must zero your sensor. Select the reading for **Force** in the bottom right corner of the window, and then select **Zero** from the menu.



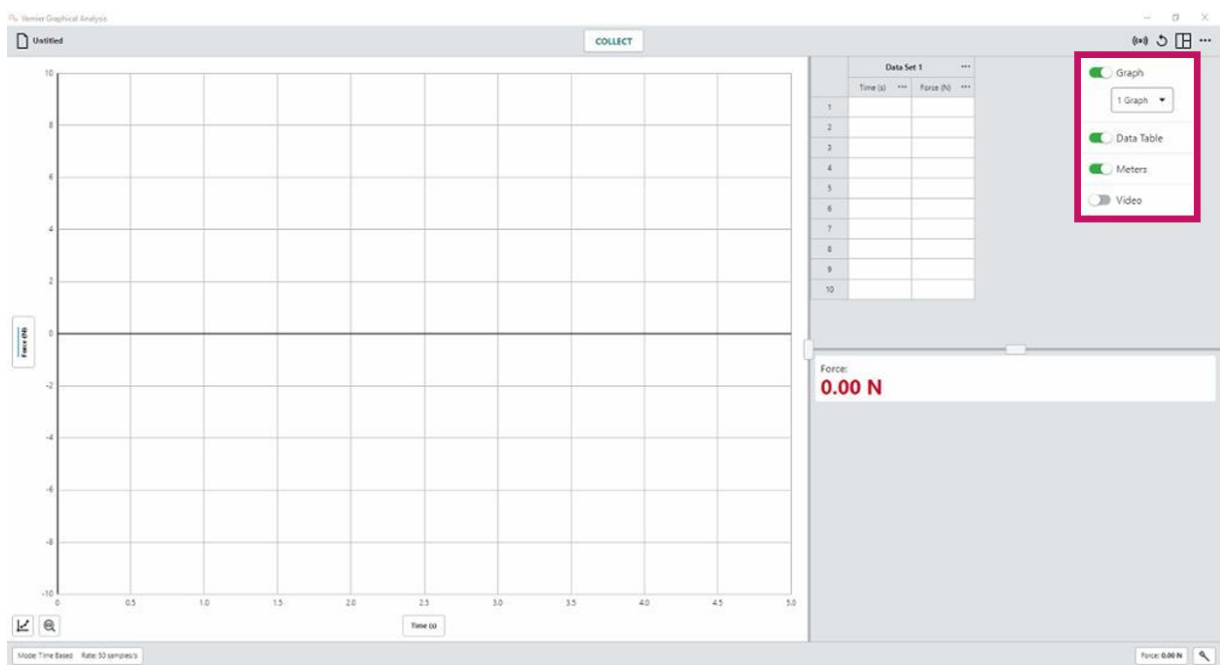
Step 3 —

In the top-right corner of the window, select the Windows option to add the Data Table and Window feature. Now you will be able to view your readings more closely.



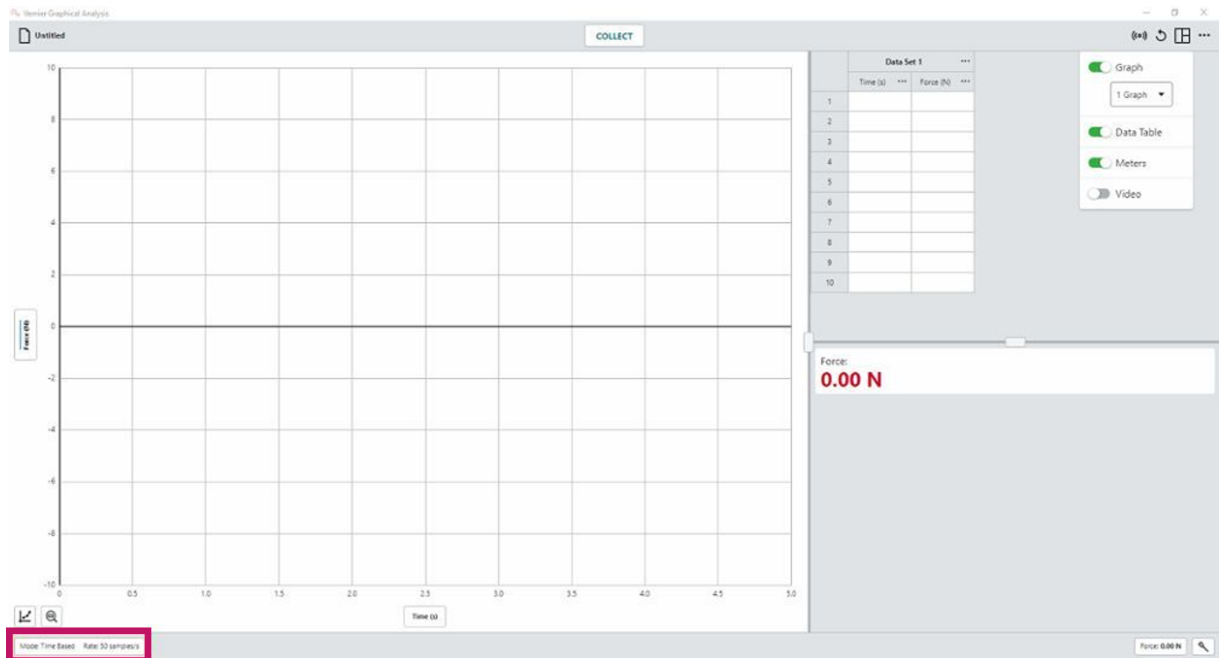
Step 4

In the right-hand pane, select **Data Table** and **Meters** to open the Data Table and Meters boxes.



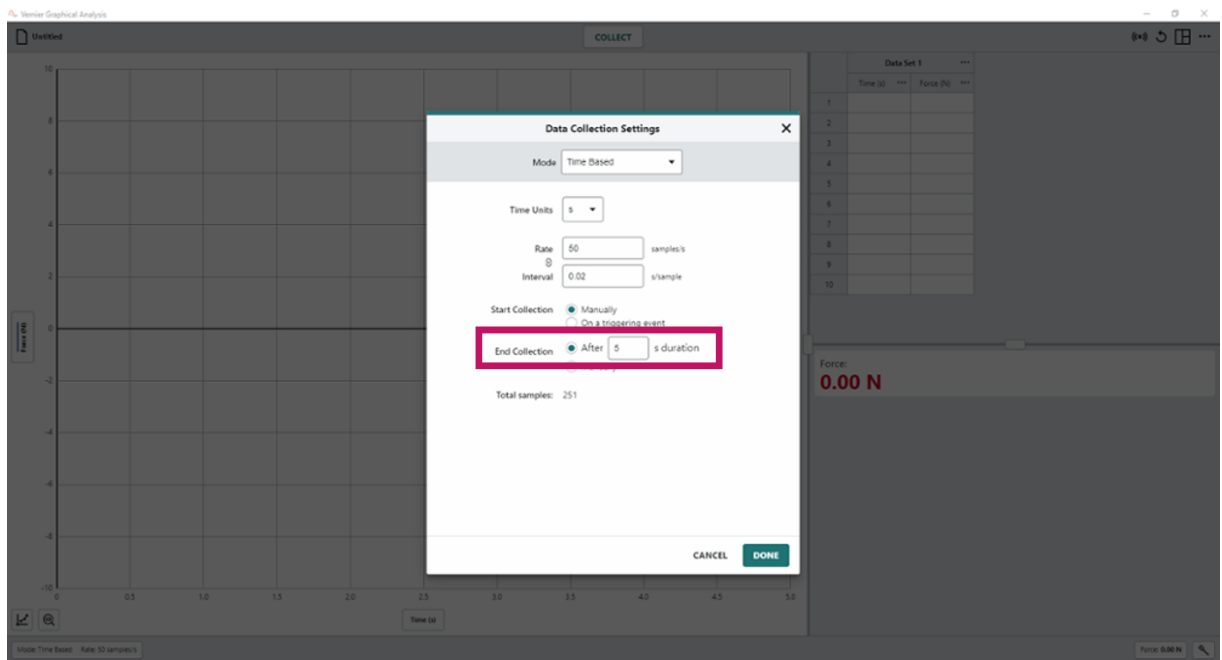
Step 5

In the bottom-left corner of the window, select **Mode** to open the Data Collection Settings box.



Step 6

In the Data Collection Settings box, set End Collection time to “After 10 s duration,” and select **DONE**.



11

Collect the data to measure the stall performance of the motor.

- Select **COLLECT** at the top of the window.
- Connect power to start the winch turning.
- Record the voltage and current displayed on the variable power supply.
- Record the force applied to the force sensor.
- Turn off the power supply and record the following values:
 - d (distance from the axis of rotation to the point where force is applied (at wheel diameter 4 inches)) and convert to meters

- $F_{\perp}$ (force to stall the motor)
- T_{stall} (stall torque) = $d \times F_{\perp} = \text{Nm}$
- Stall voltage (volts, V)
- Stall current (amps, A)

Congratulations! You have measured everything you need to characterize your motor. You have the most important design parameters that describe the motor's performance for operation at 6 V. Let's put them together to create a specification sheet, or "spec sheet" as it is commonly called.

12

In your PLTW Engineering Notebook, record the following specifications using your measured and calculated values.

- Motor model
- For operation at 6V
 - No-load current:
 - Stall current:
 - No-load speed:
 - Stall torque:

Motor Power

Engineers make many design decisions by calculating important results from these specifications. For example, an engineer may need to know how much torque is applied when a motor is operated at maximum power.

13

Calculate the torque at which the motor delivers maximum power.

Motor torque for maximum power (τ) = $\frac{1}{2}$ stall torque.

NOTE: You can review *Maximizing Motor Power* as needed.

An engineer may need to select a component based on a value calculated from the motor specifications. For example, an engineer requires a smartphone vibration motor with at least 200 mW of power.

14

Calculate the maximum output power of your motor in watts and horsepower. To find the motor output, calculate the following:

- ω (Motor speed at maximum power) = $\frac{1}{2}$ no-load speed
- P_{out} (Motor output power at maximum power) $\tau \times \omega$
- P_{out} (Motor output power at maximum power)

NOTE: The speed of the motor at half the stall torque is half the no-load speed. It is common for motor output power to be reported in horsepower (hp). The [**PLTW Engineering Formula Sheet**](#) has a conversion factor for converting power to horsepower.



Speed versus Power

Remember from *Maximizing Motor Power* that the speed versus power relationship (green line in Figure 15) is linear, and the motor speed is 0 at the stall torque.

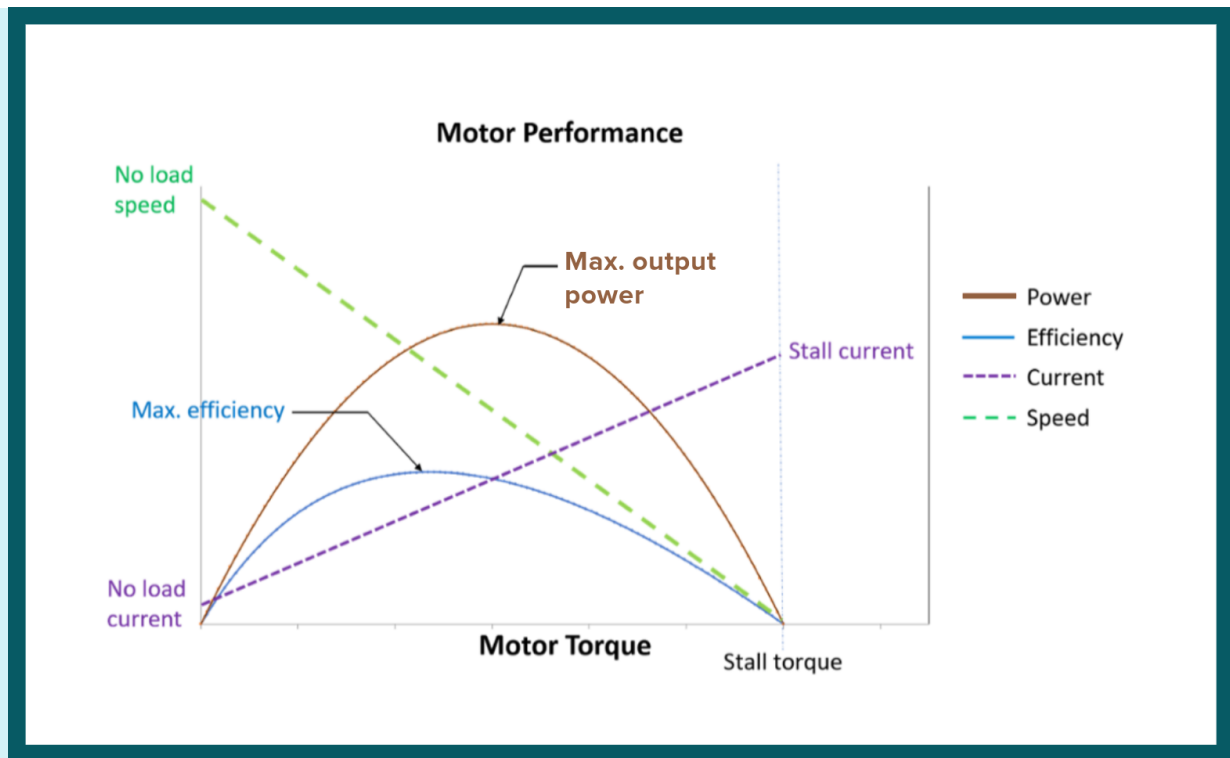


Figure 15. Motor Performance Graph

Justify that the motor speed at maximum power is half of the no-load speed.

What is the motor speed when the torque is 25% of the stall torque?

Hint

Carefully examine the x-axis, Motor Torque, and locate a point that is 25% of the way to the stall torque point. To estimate the motor speed, move from the 25% Motor Torque point vertically to the no-load speed line.

15

Research the power output of a power tool or air compressor. Create a specification sheet for the equipment you researched. Compare that tool to your motor.

16

Determine the gear ratio your motor needs to deliver its maximum power to the driveshaft.

$$GR = \frac{\tau_{out}}{\tau_{in}}$$



Additional Information

The characterization you completed for your motor can be the foundation of engineering design decisions. Consider a design challenge to drive a robot. The robot requires 10 N×m of torque to be applied on the drive axle through a gear train to transfer power from a motor to the drive axle. This means the motor will provide the input torque applied on the gear train, and the output torque from the gear train must be 10 N×m.

Maximum Motor Output Power

When the motor is exerting half of its stall torque, it is operating at maximum output power. Using a variable-current power supply (versus a variable-voltage power supply), you can vary the current to find the current halfway between the stall current and no-load current. You can use the following values, which have been calculated for your motor:

Voltage = 6.1 V

Current halfway between stall current and no-load current:

$$I_{\text{maximum output power}} = \frac{I_{\text{stall}} + I_{\text{no load}}}{2}$$

$$I_{\text{maximum output power}} = \frac{1.02 \text{ A} + 0.25 \text{ A}}{2}$$

$$I_{\text{maximum output power}} = 0.64 \text{ A}$$

Electrical power supplied to the motor:

$$P_{\text{in}} = V(I_{\text{maximum output power}})$$

$$P_{\text{in}} = 6.1 \text{ V}(0.64 \text{ A})$$

$$P_{\text{in}} = 3.9 \text{ W}$$

NOTE: The specifications for the motor you measured and calculated are the foundation for you to predict how much electrical power it will consume.

Motor Efficiency

An engineer may need to design a device that operates as efficiently as possible instead of designing it to be powerful.

17

Calculate the efficiency of your motor when it is operating at maximum power.

NOTE: For the purpose of calculating efficiency, P_{in} is the electrical power to the motor, and P_{out} is the mechanical power produced by the motor. Some energy is lost in the system. Recall the equation for efficiency:

$$\eta = \frac{P_{out}}{P_{in}} \times 100\%$$

18

With a partner, discuss and develop an [algorithm](#) to calculate the maximum efficiency at any torque.

NOTE: The calculation of torque at which the motor would deliver maximum power was rather simple (step 13). More complex was the calculation of the torque at which the motor will deliver power at maximum

efficiency. The procedure to follow for this calculation is known as an algorithm. Steps 14 and 17 guided you through the algorithm to calculate the efficiency for a given torque (half the stall torque in this example). You can generalize the process to develop an algorithm to calculate the maximum efficiency at any torque.

Automating the Motor Efficiency Calculation

To find the maximum efficiency, you can calculate the efficiency at many different torques and pick the highest efficiency. This requires many iterations of the algorithm. Computers are very good at automating that work, so a spreadsheet application with built in formulas is provided for you to create the characteristic curves of a motor at a constant voltage.



How the Spreadsheet Works

The Motor Power Spreadsheet is automated to calculate efficiencies for 100 torque values between 0 N×m and the stall torque value you enter. When you update the values in the yellow highlighted cells (Figure 16), the spreadsheet will update, specifying how the four design parameters for your motor are dependent on the torque the motor delivers.

1.00	= V = Voltage (V)		
2.000	= I_free = No-load current (A)		
3.000	= I_stall = Stall current (A)		
4.00	= omega_no-load = No-load Speed (rad/s)		
2.00	= tau_stall = Stall torque (Nm)		

Figure 16. Spreadsheet Highlighted Values

19

Open the [Motor Power Excel Spreadsheet](#) or the [Motor Power Google Sheets Spreadsheet](#) and enter the specifications of your motor. Save images of your graphs.

- With your partner, discuss these graphs, including what the x- and y-axes for each curve represent.

NOTE: Notice the two y-axes on each graph.

Color	Y-variable	Vs.	X-variable
blue	?	vs.	?
red	motor output power	vs.	motor torque
green	?	vs.	?

purple	?	vs.	?
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20

Find the point on the Efficiency graph that identifies the maximum efficiency. Indicate this spot on the graph that you saved into the spreadsheet in the previous step.

NOTE: To find the coordinates of the point of maximum efficiency, you can use mathematical representation, the data table, or the graph.

21

Answer the below prompts in your PLTW Engineering Notebook.

- Based on your graphs, what is the maximum efficiency ($\eta_{\max}$)?
- At what torque does the maximum efficiency ($T_{\max \text{ efficiency}}$) occur?
- Using the torque from the previous question as the x-coordinate on the second graph, find the current that your motor uses when it operates at maximum efficiency ($I_{\max \text{ efficiency}}$). Circle this point on your second graph.
- How much electrical power does the motor use when operating at maximum efficiency?

Reflection



The efficiency indicates the percentage of electrical energy that is transformed into mechanical energy. The rest of the energy is converted into thermal energy, increasing the temperature of the motor. Given that power is the rate at which energy is transformed, at what rate is heat energy transferred to the motor when it is operating most efficiently?

$$P_{\text{heat}} = (1 - \eta_{\text{max}})P_{\text{electrical}}$$

Conclusion

Question 1

Summarize what it means to characterize an electrical component in a system.

Question 2

What are the advantages of using math and science in engineering design? What are the advantages of trial and error?

Question 3

Explain how efficiency is related to electrical energy, mechanical energy, and heat energy.

Question 4

What gear ratio should be used for maximum power in the motor design in Figure 18?

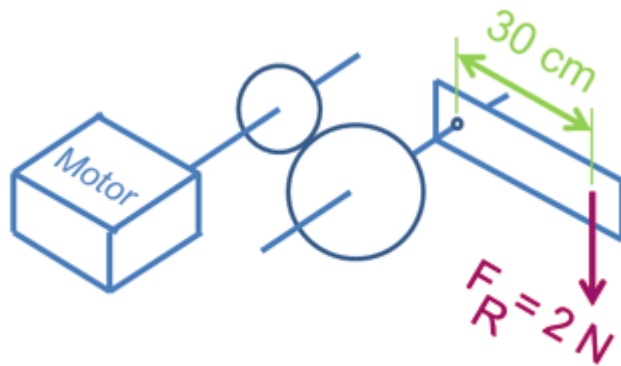


Figure 18. Motor Design and Specifications

- For operation at: 9V
- No-load current: 100mA
- Stall current: 3A
- No-load speed: 20rad/s
- Stall torque: 60 N×m